

Geometric Foundations of FFGFT

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Geometric Foundations of FFGFT

*From the Phase Condition to Circle, Polygon, Spiral, and
Nature's Preferred Ratios*

Info

Abstract. Fundamental Fractal Geometric Field Theory (FFGFT) derives all stable geometric forms from a single condition: phase coherence after a complete torus circuit. From this condition follow, in strict hierarchy, regular polygons, the circle as a limiting case, the spiral as the superposition of two frequencies, the Golden Ratio ϕ as the most stable incommensurable ratio, the Fibonacci sequence as integer approximation of ϕ , and finally the fractal coupling constant $\alpha = \phi^{-1}$. The same hierarchy appears in crystal lattices, spiral galaxies, biological growth patterns, and quantum numbers.

Cross-references: Doc. 180 (Lagrangian) · Doc. 182 (Universal scale) · Doc. 187 (TFLN photonics) · GitHub: [jpascher/T0-Time-Mass-Duality](https://github.com/jpascher/T0-Time-Mass-Duality) · Zenodo: [DOI 10.5281/zenodo.20474821](https://doi.org/10.5281/zenodo.20474821)

Contents

.1 The Universal Phase Condition

The starting point of the entire geometric hierarchy is a single condition. A field configuration Ψ can only exist stably if, after a complete circuit on the 4D torus, it agrees exactly with itself:

$$\oint_{\mathcal{T}^4} d\phi = 2\pi \cdot n, \quad n \in \mathbb{N}. \quad (1)$$

Fundamental Principle

Equation (1) is the only selection rule of FFGFT. All stable structures — geometric forms, particle masses, fundamental constants, preferred ratios — follow from this condition. Not as approximation, but as exact mathematical consequence.

The phase condition divides into three classes of solutions, which together span the complete geometric hierarchy:

1. **Discrete solutions:** $n \in \{1, 2, 3, \dots\}$ — regular polygons, quantum numbers, crystal symmetries
2. **Continuous limiting cases:** $n \rightarrow \infty$ — circle, ellipse, smooth curves
3. **Simultaneous solutions for two frequencies:** leading to incommensurable ratios — spiral, Golden Ratio, Fibonacci

.2 Level 1: Discrete Symmetries — Polygons and Crystals

Regular Polygons as Fundamental Solutions

For integer n in equation (1), a regular n -polygon emerges as the geometric realisation of the phase condition. The vertices lie at the phase angles:

$$\theta_k = \frac{2\pi k}{n}, \quad k = 0, 1, \dots, n-1. \quad (2)$$

Every regular polygon is a stable solution — a fixed point of the system. Stability increases with n , because higher modes are more densely packed and destructive interference is compensated from more directions.

The Special Role of $n = 3$: The $\mathbb{Z}_3$ Symmetry

The triangle ($n = 3$) holds a special position in FFGFT. It is the most economical stable discrete solution — three points are the minimum for a closed non-trivial configuration in the 2D cross-section of the 4D torus.

$$\mathbb{Z}_3 : \quad \theta \mapsto \theta + \frac{2\pi}{3} \quad (3)$$

The three particle generations of the Standard Model have exactly this structure. In FFGFT, $\mathbb{Z}_3$ is set, motivated by this economy and consistent with the number of generations; the generations correspond to the three $\mathbb{Z}_3$ fixed points of the torus:

$$\text{Generation } g \in \{1, 2, 3\} \longleftrightarrow \theta_g = \frac{2\pi(g-1)}{3}. \quad (4)$$

$n = 6$: Hexagonal Packing and Crystal Lattices

The hexagon ($n = 6$) is the densest packing of congruent circles in the plane — a result that follows directly from the phase condition, since $6 = 2 \times 3$ combines the $\mathbb{Z}_3$ symmetry with mirror symmetry:

$$\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2. \quad (5)$$

Honeycomb structure, graphene lattice, lithium niobate crystal lattice — all realise this symmetry on different scales.

Quantum Numbers as Polygon Orders

The principal quantum number n in atomic physics corresponds directly to the polygon order in FFGFT geometry:

$$E_n = -\frac{E_0}{n^2}, \quad n = 1, 2, 3, \dots \quad (6)$$

The $1/n^2$ spectrum itself, however, does not follow from the phase condition alone (Section .6).

.3 Level 2: The Circle as Limiting Case

Limiting case $n \rightarrow \infty$: Continuous Phase Densification

As the mode number n grows, the polygon vertices approach each other more and more densely. In the limit:

$$\lim_{n \rightarrow \infty} \left\{ \frac{2\pi k}{n} \mid k = 0, \dots, n-1 \right\} = [0, 2\pi) \quad (7)$$

the circle emerges — as continuous realisation of the phase condition. The circle is geometrically not fundamental but the limiting case of an infinite sequence of polygons.

Ellipse: Broken Symmetry

When the phase condition is fulfilled in two different spatial directions with different coupling strengths:

$$\oint_x d\phi = 2\pi n_x, \quad \oint_y d\phi = 2\pi n_y, \quad n_x \neq n_y, \quad (8)$$

an ellipse with axis ratio $n_x : n_y$ emerges. Planetary orbits, electron orbitals of higher angular momentum quantum number, resonator modes — all follow this logic.

.4 Level 3: Spirals — Superposition of Two Frequencies

Emergence of the Spiral

A spiral arises when two phase conditions must be fulfilled simultaneously: a radial and an azimuthal one. Let ω_r be the radial frequency and ω_θ the azimuthal one:

$$r(\theta) = r_0 \cdot e^{\frac{\omega_r}{\omega_\theta} \theta}. \quad (9)$$

For rational ratios $\omega_r/\omega_\theta = p/q$ the curve closes after q circuits — a Lissajous pattern. For irrational ratios it never closes — a true spiral emerges.

The Logarithmic Spiral as Self-Similarity Solution

The logarithmic spiral

$$r(\theta) = a \cdot b^\theta \quad (10)$$

is the only curve that agrees with itself under scaling:

$$r(\theta + 2\pi) = b^{2\pi} \cdot r(\theta). \quad (11)$$

It is the geometric realisation of fractal self-similarity — the core principle of FFGFT. In nature: nautilus spiral, galactic arms, phyllotaxis (leaf arrangement, sunflower seeds), hurricane structure.

.5 Level 4: The Golden Ratio ϕ as the Most Stable Incommensurable Ratio

The Problem of Two Simultaneous Phase Conditions

When a system must fulfil two simultaneous phase conditions — e.g. a radial and an azimuthal frequency — the question arises: which irrational ratio is most stable against destructive interference?

The answer from mathematics: that irrational number which is *worst* approximable by rational numbers. That is the Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} = 1.6180339 \dots \quad (12)$$

Why ϕ is optimal

ϕ has the continued fraction representation

$$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\ddots}}}$$

— all ones, the smallest possible denominators. This makes ϕ the most difficult rationally approximable number. A system that chooses ϕ as frequency ratio avoids all low rational resonances and remains stable.

Fibonacci Sequence as Integer Approximation to ϕ

The Fibonacci sequence

$$F_0 = 0, F_1 = 1, F_{k+1} = F_k + F_{k-1} \quad (13)$$

converges in its successive quotients to ϕ :

$$\lim_{k \rightarrow \infty} \frac{F_{k+1}}{F_k} = \phi. \quad (14)$$

It is the natural integer approximation to the optimal incommensurable ratio. In FFGFT it corresponds to the allowed recursion depths:

$$\mathcal{N} \in \{F_k \mid k \in \mathbb{N}\}, \quad (15)$$

since only Fibonacci numbers allow simultaneous fulfilment of two phase conditions with minimal destructive interference.

Preferred Angles: 137.5° in Nature

The Golden Angle is

$$\alpha_{\text{gold}} = 2\pi \left(1 - \frac{1}{\phi}\right) = 2\pi \cdot \phi^{-2} \approx 137.508^\circ. \quad (16)$$

A growth system that rotates successive elements by α_{gold} each time generates the densest possible packing without overlap — because no integer multiple of α_{gold} ever falls on a previously placed point. This is the reason for:

- Arrangement of leaves on a stem (phyllotaxis)
- Spiral patterns in sunflowers and pine cones
- Arrangement of scales on a pineapple

.6 Level 5: Derived FFGFT Formulae

Fractal Coupling Constant $\alpha = \phi^{-1}$

In the matrix translation (Doc. 187, Section 5), the coupling constant

$$\alpha = \phi^{-1} = \frac{2}{1 + \sqrt{5}} \approx 0.6180 \quad (17)$$

appears. It is not a free parameter but directly the reciprocal of the Golden Ratio — the scaling factor between successive fractal levels:

$$(x_{k+1}, y_{k+1}, z_{k+1}, ct_{k+1}) = \phi^{-1} \cdot (x_k, y_k, z_k, ct_k). \quad (18)$$

The $1/n^2$ Spectrum

The phase condition (1) enforces a wavelength $\lambda_n = 2\pi/n$ at n nodes. From this follows $k = n/R$ and hence $E \propto k^2 \propto n^2$. The $1/n^2$ spectrum of bound states (hydrogen, exciton, Rydberg atoms) does not follow from this and is not derived here.

No ϕ Ladder of the Lepton Masses

A ϕ -scaling ladder of the lepton masses does not hold. We have $\phi^5 = 11.090$ and $\phi^5 \pi^2 = 109.46$, far from the measured $m_\mu/m_e = 206.77 = \phi^{6.32} \pi^2$; no integer power of ϕ gives $m_\tau/m_\mu = 16.82 = \phi^{5.87}$ ($\phi^5 = 11.09$, $\phi^6 = 17.94$). A mode ladder $m_k \propto \phi^{-2k} m_0$ would give the ratio $1/2.618$ per step and falling masses. FFGFT describes the lepton masses via $m_i = r_i \xi^{p_i} v$ (Doc. 189).

Fine-Structure Constant α_{EM}

In FFGFT the fine-structure constant links the parameter ξ to the characteristic energy scale E_0 (Doc. 011):

$$\alpha_{\text{EM}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2 = \frac{4}{3} \times 10^{-4} \times (7.398)^2 = \frac{1}{137.04} \quad (19)$$

with the characteristic energy scale $E_0 = \sqrt{m_e \cdot m_\mu / K_{\text{frak}}} \approx 7.398 \text{ MeV}$, the geometric mean of the electron and muon masses (7.348 MeV), fractally corrected (Doc. 190, R72). The agreement with the measured value is a property of the anchor E_0 , which provides the only connection to the SI; it is not counted as an independent prediction.

Reference to Doc. 011

The detailed account — including the historical context (Sommerfeld) — is given in Doc. 011. Closed α formulae from ξ alone, such as a power relation $\alpha \propto \xi^{11/2}$, do not hold. This section gives only the core formula.

.7 Universal Manifestation in Nature

Same Equation, Different Scales

Table 1: Manifestation of the phase condition on different scales

System	Scale	n or ϕ^{-k}	Manifestation
Sub-Planckian	$\sim 10^{-48}$ s	$n = 1, 2, 3$	Fundamental structure ξ
Atomic orbitals	10^{-10} m	$n = 1, 2, 3, \dots$	Hydrogen spectrum
Molecules	10^{-9} m	$n = 6$	Benzene ring, DNA helix
Crystal lattices	10^{-7} m	$n = 3, 4, 6$	TFLN, graphene, honeycomb
Biology	10^{-3} m	ϕ^{-k}	Phyllotaxis, spirals
Galaxies	10^{21} m	ϕ -spiral	Spiral galaxies

The FFGFT Principle

None of these manifestations is coincidence or analogy. All solve the same equation (1) at different n and on different scales. FFGFT formalises what nature expresses at all levels: stable structures arise where phase coherence is guaranteed.

Self-Organisation as Geometric Necessity

Self-organisation — as observed in TFLN crystals, biological growth, or the formation of galactic structures — is not an emergent property in FFGFT that needs to be explained. It is geometrically necessary: every system drifts into the fixed points of the recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$, because no other stable states exist. The apparent “intelligence” of nature in finding forms is the phase condition (1).

.8 Self-Organisation Below the Wavelength Limit: Block Copolymers in Chip Manufacturing

The Physical Limit of Lithography

Modern chip manufacturing encounters a fundamental physical limit: structures can only be directly “written” with light if their dimensions are larger than half the wavelength of the light used. Even with extreme UV lithography (EUVL, $\lambda \approx 13.5$ nm), structures below ~ 10 nm are no longer directly exposable — diffraction and stochastic defects prevent it [5, 6].

Directed Self-Assembly: Structures without Direct Writing

The solution is a paradigm shift from “top-down” to “bottom-up”: **Directed Self-Assembly (DSA)** of block copolymers (BCPs).

A block copolymer consists of two chemically distinct polymer chains covalently linked. Since chemically dissimilar blocks repel each other, the chains spontaneously separate

into periodic nanophases — without each individual structure having to be externally programmed:

- **Periodicities:** 5–50 nm, well below the EUV wavelength
- **Control parameter temperature:** heating above the glass transition temperature activates phase separation; the temperature determines which mode (lamellar, cylindrical, spherical) is preferred
- **Control parameter substrate/stress:** chemical or geometric prestructuring of the substrate “presses” the modes into prescribed paths (guided DSA)

A recent example: at a template pitch of 150 nm, DSA produced a hexagonal hole structure with a pitch of 50 nm — a density tripling compared to the optically exposed pattern.

Crucial Observation

The material finds structures *smaller* than the wavelength of the light used — not because the light creates these structures, but because the system spontaneously drifts into geometrically stable states. The external excitation scale (EUV, 13 nm) and the resulting structure scale (5–10 nm) need not coincide.

FFGFT Interpretation: Scale-Invariant Fixed Points

This behaviour is no surprise in FFGFT but geometrically necessary. The phase condition (1) has stable solutions at discrete mode numbers n with energies E_n . The block copolymer locks into that mode n^* which, at given temperature T and substrate boundary condition U , is energetically minimal:

$$n^*(T, U) = \arg \min_n [E_n - \mu(T) \cdot n - \lambda(U) \cdot n^2]. \quad (20)$$

Here $\mu(T)$ is the thermal contribution (temperature selects the mode) and $\lambda(U)$ the substrate geometry contribution (guided DSA enforces specific winding numbers). The resulting structure is *stable* because destructive interference suppresses all other modes — exactly the lock-in mechanism of the 4D torus.

The FFGFT Principle of Sub-Wavelength Lock-In

A system need not lock in on the scale of its excitation. The recursion $\Psi(t) = G(\Psi(t - T_{\text{rev}}), \xi)$ has fixed points on all scales — the realised fixed point depends on boundary conditions, not on the excitation wavelength. Block copolymers under EUV exposure show: the system finds structures far below the light wavelength because the geometric condition (1) is scale-invariant.

Connection to the Universal Form Hierarchy

The nano-patterns emerging from DSA — lamellae, cylinders, spheres, gyroids — are exactly the geometric basic forms from Section 2: $n = 2$ (lamellar, stripe pattern), $n = 3$ (hexagonal-cylindrical), $n = 6$ (cubic-spherical). The material does not choose randomly — it solves the same phase condition that, on any other scale, produces polygons, crystal lattices, and quantum numbers.

.9 Conclusion

- A single equation — $\oint d\phi = 2\pi n$ — is the origin of all geometric forms and preferred ratios.
- Regular polygons, circle, ellipse, and spiral are different solution classes of the same condition.
- The Golden Ratio ϕ is the most stable incommensurable frequency ratio — mathematically inevitable, not empirical.
- The Fibonacci sequence is the integer approximation of ϕ and defines the allowed recursion depths in FFGFT.
- The fractal coupling constant $\alpha = \phi^{-1}$ follows directly from this geometric hierarchy. The $1/n^2$ spectrum is not derived here, a ϕ ladder of the lepton masses does not hold, and the agreement of α_{EM} with the measured value is a property of the anchor E_0 .
- The same hierarchy appears on all scales — from the sub-Planckian fundamental structure to spiral galaxies.

Core Statement

Nature is not diverse despite a single fundamental structure. Nature is diverse *because* a single geometric condition has infinitely many solutions on infinitely many scales. FFGFT is the formalisation of this fact.

Bibliography

Bibliography

- [1] J. Pascher, *Lagrangian derivation of FFGFT*, T0-Doc. 180 (2026).
- [2] J. Pascher, *Universal scale and cosmological limits in FFGFT*, T0-Doc. 182 (2026).
- [3] J. Pascher, *FFGFT Analytics: Photonics, matrix space and B-meson anomalies*, T0-Doc. 187 (2026).
- [4] M. Livio, *The Golden Ratio: The Story of Phi, the World's Most Astonishing Number*, Broadway Books (2002).
- [5] Review: *Directed block copolymer self-assembly for next-generation lithography*, ScienceDirect (2026). [doi:10.1016/j.device.2026.100583](https://doi.org/10.1016/j.device.2026.100583)
- [6] K. Lee, K. Kim, E. Vargo et al. (U.Chicago, Berkeley Lab, Argonne), *Directed self-assembly of block copolymers for high-precision patterning in the era of extreme ultraviolet lithography*, MRS Communications (October 2025). [doi:10.1557/s43579-025-00841-7](https://doi.org/10.1557/s43579-025-00841-7)
- [7] J. Zhan et al. (Fudan University, Zhangjiang Laboratory), *Directed Self-Assembly of an Acid-Responsive Block Copolymer for Hole-Shrink Process and Pattern Transfer*, Nanomaterials, Vol. 15 (2025). [doi:10.3390/nano15201571](https://doi.org/10.3390/nano15201571)
- [8] P. Prusinkiewicz, A. Lindenmayer, *The Algorithmic Beauty of Plants*, Springer (1990).